

1. 70–90% kids, 10–15% adults

WHAT YOU SEE

Gate: 70-90% KIDS, 10-15% ADULTS

WHAT IT ENCODES

Minimal change disease (MCD) is the most common cause of nephrotic syndrome in children (70–90% of pediatric nephrotic syndrome) and a minority (~10–15%) in adults. It presents as abrupt, often dramatic edema in an otherwise well child.

BOARD TRAP

Childhood nephrotic syndrome is MCD until proven otherwise; FSGS becomes more likely if steroid-resistant or if there is hematuria/hypertension/renal insufficiency.

2. LM normal / EM only

WHAT YOU SEE

Microscope: LM normal, NIL lesion; magnifier EM ONLY

WHAT IT ENCODES

Light microscopy is essentially normal ('nil disease') and immunofluorescence is negative. The diagnosis is made on electron microscopy, which shows diffuse podocyte foot-process effacement — the sole structural lesion.

BOARD TRAP

Normal LM with negative IF doesn't mean no disease — EM effacement defines MCD; you cannot diagnose it on LM/IF alone.

3. Foot-process effacement

WHAT YOU SEE

Sandbox labeled FOOT PROCESS EFFACEMENT

WHAT IT ENCODES

EM shows diffuse effacement (fusion) of podocyte foot processes with no immune deposits. This loss of the slit-pore architecture causes the selective protein leak. A circulating T-cell-derived permeability factor is implicated.

BOARD TRAP

Effacement is shared with FSGS; in MCD it is diffuse WITHOUT segmental sclerosis — sampling that misses sclerosis can mimic MCD (really FSGS).

4. Selective proteinuria (albumin)

WHAT YOU SEE

Albumin balloons; SELECTIVE PROTEINURIA, 10 g/24h

WHAT IT ENCODES

MCD causes highly selective proteinuria — predominantly albumin — because the charge barrier is lost while size selectivity is preserved. Proteinuria can be massive (>10 g/day) with profound hypoalbuminemia and edema.

BOARD TRAP

Selective (mostly albumin) loss favors MCD; nonselective proteinuria with larger proteins suggests FSGS or membranous instead.

5. Slit-pore membrane / CD80

WHAT YOU SEE

SLIT-PORE MEMBRANE label; CD80 (nephrin) markers

WHAT IT ENCODES

Podocyte slit-diaphragm proteins (nephrin) and immune mediators (CD80/B7-1) are implicated in the charge-barrier disruption. Anti-nephrin antibodies are an emerging marker in some MCD.

BOARD TRAP

MCD is a podocytopathy of the slit diaphragm without immune-complex deposits — IF is negative, unlike membranous or lupus.

6. 1st line: steroids, 8 wks

WHAT YOU SEE

Trophy: 1ST LINE, 8 WEEKS, 90% kids, 80% adults respond

WHAT IT ENCODES

High-dose corticosteroids are first-line and induce remission in ~90% of children and ~80% of adults. Children typically respond within weeks; adults respond more slowly and need longer courses.

BOARD TRAP

Steroid responsiveness is so high in children that empiric steroids are given first — biopsy is reserved for atypical or steroid-resistant cases.

7. No biopsy first (kids)

WHAT YOU SEE

Sign: NO BIOPSY FIRST

WHAT IT ENCODES

In a typical child with nephrotic syndrome, empiric steroids are started without biopsy because MCD is overwhelmingly likely and steroid-responsive. Biopsy is done if steroid-resistant, frequently relapsing, or atypical features are present.

BOARD TRAP

Adults usually DO get a biopsy first because MCD is less common and the differential (FSGS, membranous) is broader.

8. Relapse / steroid-dependent

WHAT YOU SEE

Merry-go-round: RELAPSE; steroid-dependent, frequent relapse, steroid-resistant

WHAT IT ENCODES

Many children relapse; frequently-relapsing or steroid-dependent disease warrants steroid-sparing agents. Persistent steroid resistance should prompt biopsy to reconsider FSGS.

BOARD TRAP

Steroid resistance in presumed MCD should raise suspicion for FSGS on rebiopsy — the lesion may have been missed by sampling.

9. 2nd line agents

WHAT YOU SEE

2ND LINE shelf: cyclophosphamide, CNI, MMF, rituximab

WHAT IT ENCODES

For frequently relapsing, steroid-dependent, or steroid-resistant MCD, steroid-sparing options include calcineurin inhibitors, cyclophosphamide, mycophenolate (MMF), and rituximab. Rituximab is increasingly used for relapsing disease.

BOARD TRAP

Second-line agents are for relapsing/steroid-dependent disease — they are not first-line; steroids come first in typical MCD.

10. Associations: NSAIDs, Hodgkin, lithium

WHAT YOU SEE

ASSOCIATIONS board: NSAIDs, Hodgkin lymphoma, lithium

WHAT IT ENCODES

Secondary MCD is classically associated with NSAIDs (often with concurrent acute interstitial nephritis), Hodgkin lymphoma, and lithium. Atopy is also linked.

BOARD TRAP

MCD plus AIN points to NSAIDs; MCD with B-symptoms/lymphadenopathy points to Hodgkin lymphoma — look for the trigger in adults.

11. Diuretic-responsive edema

WHAT YOU SEE

Anasarca patient; diuretic-responsive, hypovolemia risk

WHAT IT ENCODES

Edema is often diuretic-responsive, but aggressive diuresis risks intravascular volume depletion and AKI because oncotic pressure is already low. Cautious diuresis with albumin in select cases.

BOARD TRAP

Over-diuresing a hypoalbuminemic nephrotic patient can precipitate prerenal AKI — the intravascular space is underfilled despite total-body edema.

12. Bland sediment

WHAT YOU SEE

BLAND SEDIMENT label; anti-nephtrin

WHAT IT ENCODES

Urinalysis shows a bland sediment — no significant hematuria, no RBC casts — consistent with a pure podocytopathy and nephrotic (not nephritic) physiology.

BOARD TRAP

Active sediment with RBC casts argues against MCD and toward a nephritic process; MCD sediment is bland.

13. Steroid-resistant → go deeper (FSGS)

WHAT YOU SEE

Sign: STEROID RESISTANT? GO DEEPER; FSGS, atopy 40%

WHAT IT ENCODES

Steroid resistance should prompt rebiopsy and consideration of FSGS, which shares foot-process effacement but adds segmental sclerosis and carries a worse prognosis. MCD and FSGS may lie on a spectrum.

BOARD TRAP

Don't keep escalating steroids in resistant 'MCD' — reclassify as FSGS, which needs longer therapy/CNI and has worse outcomes.
